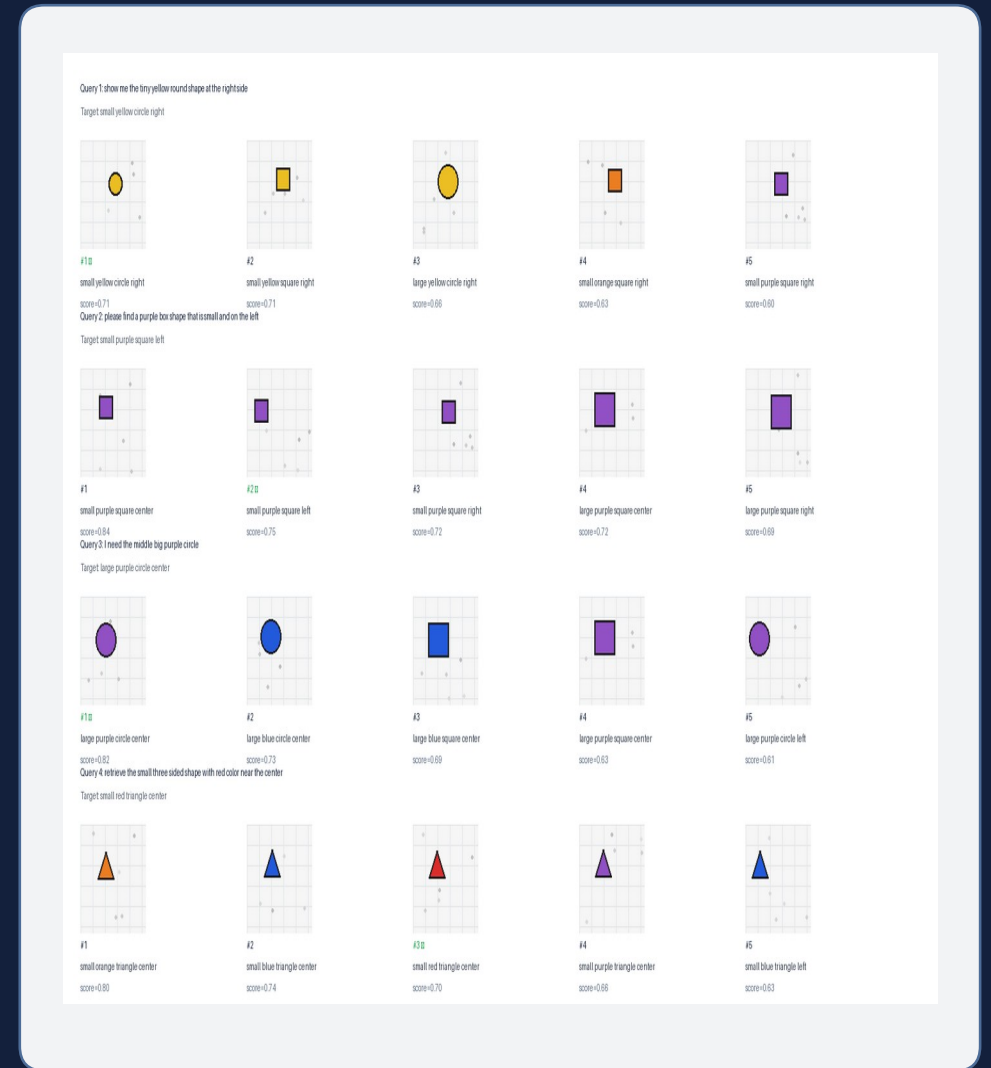


Mini Research Problem

Text-to-Image Search with a CLIP-style System

Redo version: stronger baseline, harder language queries, visually similar distractors, and actual run results inside the deck



Course directions used: Multi-Modality + Transformers + Computer Vision + Optimization

TL;DR: retrieval result

Motivation

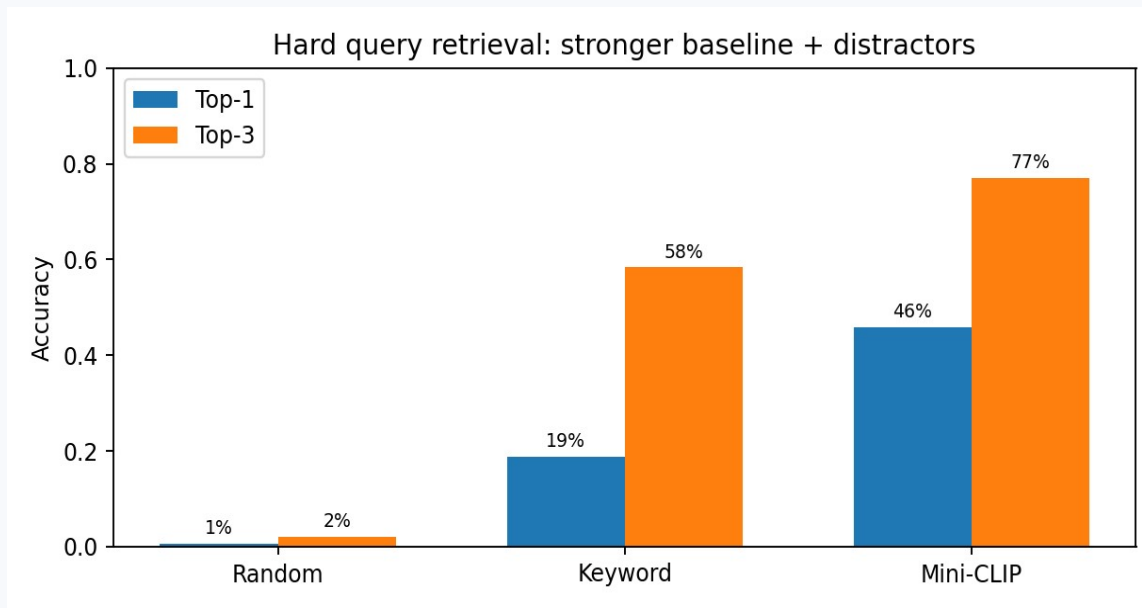
Text-to-image search should support flexible natural language, not only exact labels or .filenames

Main idea

A CLIP-style dual encoder maps images and text into one shared embedding space, then ranks .images by similarity

Actual result

On hard queries: Mini-CLIP reached 45.8% Top-1 and .77.1% Top-3, beating keyword search (18.8% / 58.3%)



Method	Top-1	Top-3	Top-5
Random	0.7%	2.1%	3.5%
Keyword	18.8%	58.3%	81.3%
Mini-CLIP	45.8%	77.1%	87.5%

Dataset: 144 images, 576 image-text training pairs, 48 hard .evaluation queries
Benchmark is no longer exact-caption retrieval: it includes .synonyms and close distractors

Problem: searching images with language is not just classification

.The input and output live in different modalities: text query in, image ranking out

Why normal CV is limited

A classifier predicts from fixed labels. But a user may search for “tiny blue box on the right”, “round object in the middle”, or “three sided red shape”. These are .compositional queries

What retrieval needs

The system needs to understand multiple attributes at once: object type, color, size, and position. It also must rank many similar images, not only output one .class label

Research question

Can a small CLIP-style model learn a shared image-text representation that beats keyword
?search under harder queries and distractors

Core task definition

.Given a text query q and image collection $I = \{i_1, i_2, \dots\}$, encode q and each image into vectors, compute cosine similarity, and return the top- k nearest images

Literature review: related methods

.The project sits in vision-language representation learning

Method	Main idea	Relation to this project
Keyword / metadata	.Search captions, tags, filenames, or labels	.Simple baseline; fails when wording differs from metadata
CLIP	.Contrastive Language-Image Pre-training with image and text encoders	.Main model family for this project: shared embedding retrieval
ALIGN	.Large-scale image-text contrastive learning on noisy web pairs	.Similar to CLIP but emphasizes scale and noisy data
BLIP	.Bootstrapping Language-Image Pre-training for understanding and generation	.Useful for captioning/VQA; more complex than needed here

Links used: CLIP paper <https://arxiv.org/abs/2103.00020> | OpenAI CLIP <https://github.com/openai/CLIP> | ALIGN <https://arxiv.org/abs/2102.05918> | BLIP <https://arxiv.org/abs/2201.12086> | BLIP code <https://github.com/salesforce/BLIP>

Approach: CLIP-style dual encoder architecture

.The architecture is designed around one shared embedding space

Mini-CLIP architecture used in the built system



Image encoder: a small CNN extracts visual features
from the generated shape image

Text encoder: token embeddings are pooled and
projected into the same 64-D space

Optimization: symmetric contrastive loss pulls
matched image-text pairs together and pushes
mismatches apart

Retrieval: image vectors are precomputed; a query
vector is compared with cosine similarity

Important architecture

.Because queries ask about left/center/right, the CNN keeps its spatial feature map before projection. Global pooling would lose too much position information

Built system: what the code actually does

.The code generated all result images used in the deck

Generate 144 synthetic images from color x shape x position x size

.combinations

Create 576 training captions with multiple natural-language templates

.per image

Train Mini-CLIP for 18 epochs using AdamW and symmetric contrastive

.cross-entropy

Build an image embedding index and evaluate 48 hard natural-

.language queries

Compare Random expected baseline, Keyword baseline, and Mini-CLIP

.retrieval

```
QUERY: show me the tiny yellow round shape at the right side
TARGET: small yellow circle right
small orange square right | score=0.711 .1
small yellow square right | score=0.706 .2
small yellow circle right | score=0.664 <== target .3
small purple square right | score=0.619 .4
large orange circle right | score=0.555 .5
QUERY: please find a purple box shape that is small and on the left
TARGET: small purple square left
small purple square center | score=0.753 .1
small purple square left | score=0.678 <== target .2
large purple square center | score=0.642 .3
small purple square right | score=0.590 .4
large purple square left | score=0.573 .5
QUERY: I need the middle big purple circle
TARGET: large purple circle center
large blue circle center | score=0.821 .1
large purple circle center | score=0.784 <== target .2
large blue square center | score=0.734 .3
small blue square center | score=0.689 .4
large orange circle center | score=0.614 .5
QUERY: retrieve the small three sided shape with red color near the center
TARGET: small red triangle center
small orange triangle center | score=0.813 .1
small red triangle center | score=0.775 <== target .2
small blue triangle center | score=0.743 .3
small purple triangle center | score=0.642 .4
small blue triangle left | score=0.625 .5
QUERY: show me the tiny green five point shape at the left side
TARGET: small green star left
small green star left | score=0.827 <== target .1
large purple star left | score=0.777 .2
large green star left | score=0.775 .3
small purple star left | score=0.772 .4
small green triangle left | score=0.705 .5
QUERY: please find a yellow five point shape that is large and on the right
TARGET: large yellow star right
large orange square right | score=0.624 .1
large yellow square right | score=0.618 .2
large yellow circle right | score=0.606 .3
large orange circle right | score=0.577 .4
small orange square right | score=0.522 .5
```

Design: Test the ability of the system with inputs

Hard queries .1

Evaluation uses natural phrasing and synonyms: "tiny", "big", "box shape", "round shape", "three sided shape", "middle", and "right side"

Similar distractors .2

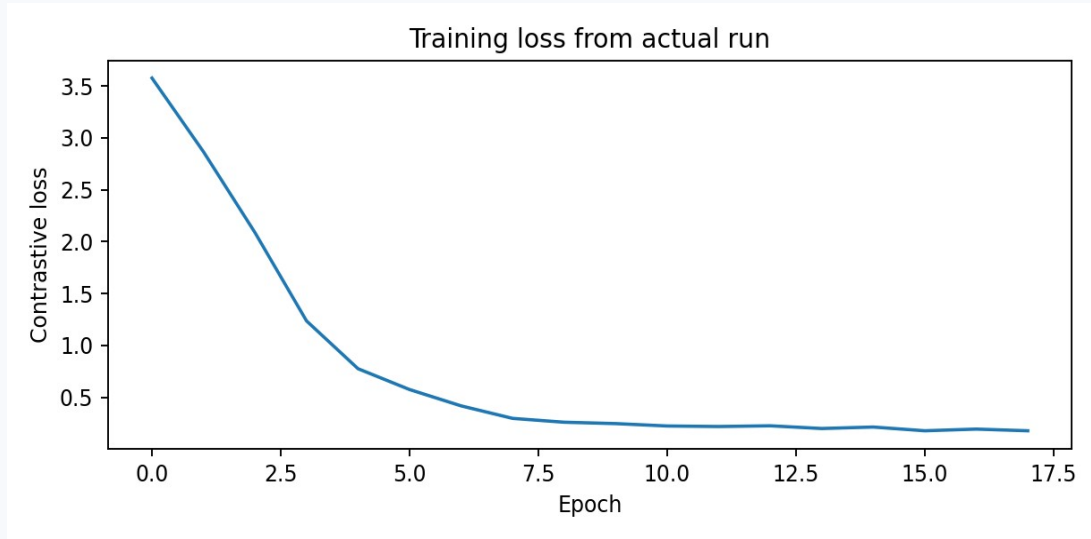
The index keeps all other shape/color/size/position combinations, so many wrong images differ by only one attribute

Why this test the ability

Real retrieval systems usually do not receive exact captions. They handle user wording, ambiguous language, and many near-duplicate or visually similar candidates. That is why Top-3 and Top-5 are also reported

Training and evaluation setup

.The evaluation is intentionally more difficult than exact-caption lookup



Final loss after 18 epochs: 0.176 | embedding dimension: 64

Training data

images and 576 image-text pairs. Captions include alternative wording for 144 .each image

Hard benchmark

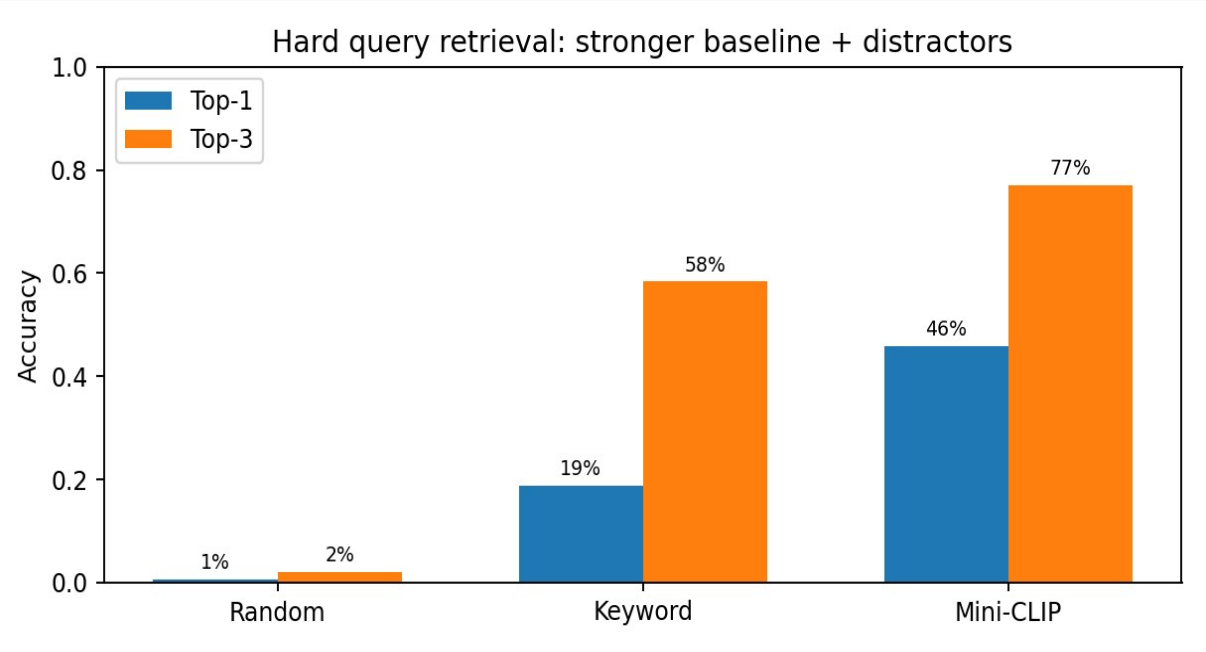
queries with synonyms and natural phrasing. Search index includes all 48 .distractor combinations

Metrics

Top-1, Top-3, Top-5, MRR, median rank. Top-k is important because retrieval .systems usually show a ranked list

Run results: better than keyword

.The result now shows a realistic gain instead of a near 0% to 100% jump



Method	Top-1	Top-3	MRR
Random expected	0.7%	2.1%	0.039
Keyword baseline	18.8%	58.3%	0.441
Mini-CLIP	45.8%	77.1%	0.634

Interpretation

Keyword search is already reasonable because the dataset has captions. Mini-CLIP still improves because it learns synonym/attribute alignment and ranks by visual-text similarity

Qualitative results: successes and misses

.The top results are usually semantically close, even when Top-1 is not always the exact target

Query 1: show me the tiny yellow round shape at the rightside
Target: small yellow circle right

#1 
small yellow circle right
score=0.71

Query 2: please find a purple boxshape that is small and on the left
Target: small purple square left

#1 
small purple square center
score=0.84

Query 3: I need the middle big purple circle
Target: large purple circle center

#1 
large purple circle center
score=0.82

Query 4: retrieve the small three sided shape with red color near the center
Target: small red triangle center

#1 
small orange triangle center
score=0.80

#2 
small yellow square right
score=0.71

#2 
small purple square left
score=0.75

#2 
large blue circle center
score=0.73

#2 
small blue triangle center
score=0.74

#3 
large yellow circle right
score=0.66

#3 
small purple square right
score=0.72

#3 
large blue square center
score=0.69

#3 
small red triangle center
score=0.70

#4 
small orange square right
score=0.63

#4 
large purple square center
score=0.72

#4 
large purple square center
score=0.63

#4 
small purple triangle center
score=0.66

#5 
small purple square right
score=0.60

#5 
large purple square right
score=0.69

#5 
large purple circle left
score=0.61

#5 
small blue triangle left
score=0.63

Notice the distractors: wrong results often share most attributes with the target, such as same color and shape but different size or position. That is the realistic failure mode created in this redo

Required links and deliverables

Code link

<https://github.com/Keming-Z/ai-hw-spring-2026-final/tree/master/code>

Relevant method links

CLIP paper, OpenAI CLIP repo, ALIGN paper, BLIP paper, and BLIP repo are listed in the literature slide

Conclusion

- .Text-to-image search is a multi-modality problem: the model must align language and visual features
- .The CLIP-style dual encoder is efficient because image embeddings can be precomputed and reused
- .Contrastive optimization is the core approach: match positive image-text pairs and separate negatives
- .The redo benchmark is harder: stronger keyword baseline, hard natural-language queries, and similar distractors
- .Actual results show Mini-CLIP improves retrieval without pretending the system is perfect

Final takeaway

- .Mini-CLIP improves from keyword Top-1 18.8% to 45.8%, and from keyword Top-3 58.3% to 77.1% on the harder benchmark

This connects directly to course topics: Transformers, Computer Vision, Multi-Modality, Representation Learning, and .neural network optimization